
EquiCAD: A Geometric Equivariant Neural Network for 3D Shape Classification

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Abstract

Three-dimensional (3D) shape classification plays a central role in computer vision and computer-aided design (CAD), underpinning applications in intelligent manufacturing, automated inspection, and digital engineering. Despite recent progress with 3D CNNs and graph-based approaches, existing methods often overlook the geometric-topological regularities and symmetry principles intrinsic to CAD boundary representations (B-reps). To address this challenge, we introduce EquiCAD, a symmetry-aware learning framework that integrates equivariant representations with graph-based reasoning. By leveraging group-theoretic decomposition of curve and surface descriptors, EquiCAD preserves symmetry-consistent feature transformations while retaining rich geometric details. The model further exploits hierarchical message passing to capture interactions between local features and global structure. Experimental results across multiple datasets, including SolidLetters, Parts, the Machining Feature benchmark, and our newly constructed Features dataset, demonstrate substantial improvements over prior state-of-the-art approaches, particularly on industrially relevant shapes with fine-grained attributes. These findings highlight the value of symmetry-aware modeling for robust and generalizable 3D shape analysis.

1. Introduction

Three-dimensional shape classification is a cornerstone problem in computer vision and geometric deep learning, supported by large-scale shape repositories, neural shape representations, and CAD reverse-engineering tools

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(Chang et al., 2015; Park et al., 2019; Liu et al., 2024). Among various 3D representations, Boundary Representation (B-rep) CAD models are particularly significant in manufacturing and engineering, as they offer a precise, compact, and feature-rich description of solid objects (Lambourne et al., 2021; Jayaraman et al., 2021; Zhang et al., 2024). However, the complex, structured nature of B-rep data—comprising vertices, edges, and faces along with their intricate topological relationships—presents a formidable challenge for automated recognition systems. The task becomes especially difficult when aiming for fine-grained geometric perception that is not only accurate but also robust to rigid transformations such as rotations and reflections, a fundamental requirement for real-world deployment (Thomas et al., 2018; Fuchs et al., 2020; Geiger & Smidt, 2022). These developments are part of a broader movement toward AI-assisted engineering systems, ranging from quantitatively constrained CAD generation (Zheng et al., 2025) to energy- and QoS-aware optimization in UAV-assisted vehicular edge computing (Dai et al., 2026); our work focuses on reliable geometric understanding of B-rep CAD models.

Early attempts at CAD shape analysis often relied on hand-crafted, rule-based methods that leverage domain-specific geometric heuristics. While interpretable, these approaches are inherently limited in scalability and generalization, as they require extensive manual feature engineering and struggle to adapt to complex or unseen shape variations. The advent of data-driven techniques broadened 3D learning to point sets and neural implicit fields, which are compatible with standard neural architectures but do not explicitly encode B-rep topology (Qi et al., 2017a; Park et al., 2019; Mildenhall et al., 2020). Such representations can discard the rich topological structure and parametric geometry of B-reps, leading to suboptimal performance on fine-grained, feature-aware tasks. Moreover, they are often sensitive to sampling density and fail to preserve continuity and symmetry properties, limiting their applicability to high-precision CAD analysis.

More recent B-rep-oriented methods (Lambourne et al., 2021; Wu et al., 2024; Li et al., 2025; Park et al., 2025) have sought to model CAD topology and geometry, often through graph or adjacency-based representations of faces,

edges, and their relationships. Although these approaches better preserve structural information, they typically overlook the fundamental role of symmetry. Standard graph neural networks (GNNs) applied to B-rep graphs are not inherently equivariant to 3D rotations and reflections, causing them to learn pose-dependent features that hamper robustness and generalization (Bronstein et al., 2017; Cohen & Welling, 2017). Furthermore, many GNN-based models do not explicitly account for the heterogeneous nature of B-rep entities (e.g., faces, edges, loops) or the multi-level interactions between local geometric descriptors and global shape characteristics. This often results in limited expressiveness for capturing hierarchical part-whole relationships and symmetry-aware shape semantics.

To overcome these limitations, we introduce EquiCAD, a symmetry-aware learning framework that integrates group-theoretic equivariance with hierarchical graph-based reasoning for B-rep classification. Our method systematically unifies $SO(3)/O(3)$ -equivariant neural networks with graph representation learning on CAD data (Cohen & Welling, 2017; Thomas et al., 2018). EquiCAD introduces a symmetry-preserving encoding scheme that decomposes geometric entities from B-reps into irreducible representations based on physical parity, and then converts equivariant features into invariant embeddings before fusion and graph-level classification. Additionally, the framework employs hierarchical message passing to capture multi-scale interactions between local features and global structure. Extensive experiments on four benchmarks demonstrate that EquiCAD achieves state-of-the-art performance, underscoring the critical role of symmetry-aware modeling in advancing 3D shape understanding.

Our main contributions are summarized as follows:

- We introduce EquiCAD, the first framework that systematically unifies $SO(3)/O(3)$ -equivariant neural networks with graph neural networks for 3D solid model classification.
- We propose a symmetry-preserving encoding scheme that assigns B-rep entities to irreducible representations according to physical parity and converts the resulting equivariant features into $SO(3)/O(3)$ -invariant embeddings before fusion and classification.
- We curate a new benchmarking dataset, featuring 16 challenging sub-tasks selected from 24 binary feature-recognition tasks in CAD models, to facilitate rigorous evaluation of equivariance-based methods.
- Through extensive experiments on four benchmarks, we demonstrate that EquiCAD significantly outperforms state-of-the-art baselines, validating the importance of symmetry priors for robust and generalizable 3D shape classification.

2. Related Work

3D Shape Classification with Deep Learning. The advent of deep learning has spurred diverse approaches to 3D shape understanding. Large-scale shape repositories and neural implicit representations have expanded the scope of 3D learning beyond handcrafted descriptors (Chang et al., 2015; Park et al., 2019; Mildenhall et al., 2020). Point-based architectures, such as PointNet (Qi et al., 2017a) and its successors (Qi et al., 2017b; Wang et al., 2019), revolutionized the field by directly consuming unordered point clouds. While these methods achieve permutation invariance, they are inherently designed for unstructured data and do not capitalize on the rich, structured geometric and topological information inherent to CAD boundary representations (B-reps) (Bronstein et al., 2017; Guo et al., 2021), which is critical for industrial applications.

Neural Networks for CAD Data. To better exploit the structure of CAD models, recent research has focused on representations that natively encode B-rep data. UV-Net (Jayaraman et al., 2021) pioneered a hybrid architecture that processes individual B-rep faces using 2D CNNs on their parameter space (UV patches) and models their topological connectivity via a graph neural network (GNN). This approach demonstrated superior performance on CAD-specific tasks compared to point- or voxel-based methods. Related CAD learning pipelines include generative CAD sequence modeling (Wu et al., 2021), B-rep graph and feature-recognition models (Lambourne et al., 2021; Zhang et al., 2024; Park et al., 2025), and quantitatively constrained CAD generation (Zheng et al., 2025).

$SO(3)/O(3)$ -Equivariant Deep Learning. Symmetry-aware geometric neural networks have proved effective in physical and geometric domains, from invariant molecular modeling (Schütt et al., 2017) to steerable and equivariant architectures. Equivariant neural networks provide a principled framework for building invariance or equivariance to specific symmetry groups (e.g., $E(3)$, $SO(3)$, $O(3)$) directly into the model architecture (Weiler et al., 2018; Fuchs et al., 2020). By leveraging steerable features and irreducible representations (irreps), models such as those built with the e3nn framework (Geiger & Smidt, 2022) have shown remarkable data efficiency and generalization. However, the application of these powerful principles has been largely confined to relatively simple 3D data types like point clouds (e.g., Vector Neurons (Deng et al., 2021)), spherical signals, or voxels.

Hybrid and Multi-Modal Architectures. Bridging different neural network paradigms has proven effective for tackling complex data. Architectures that combine local geometric encoders with relational or topology-aware modules (Hanocka et al., 2019; Wu et al., 2024; Li et al., 2025) can capture both local patterns and structural dependen-

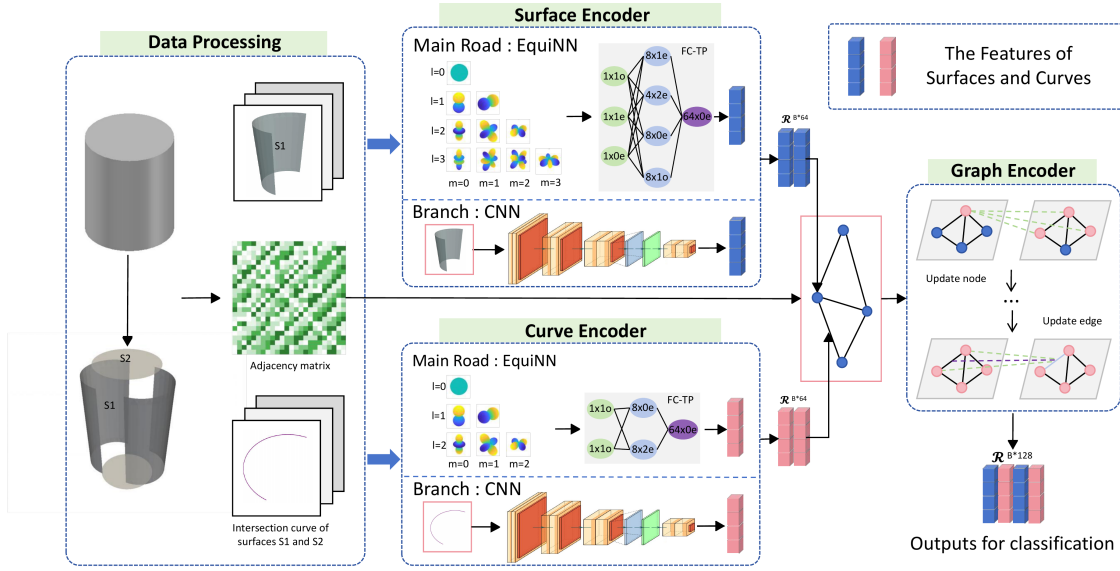


Figure 1. Overview of the proposed EquiCAD framework. Our approach extracts meaningful features of B-rep models through three synergistic steps: (1) an $SO(3)/O(3)$ -equivariant encoder that preserves geometric symmetries, (2) an auxiliary CNN branch operating on invariantized equivariant features to capture local UV-space patterns, and (3) a graph neural network that aggregates topological relationships. Feature fusion combines invariant embeddings before message passing produces the final graph-level embedding for classification.

cies. Similarly, recent efforts have begun to integrate equivariant layers with standard architectures (Thomas et al., 2018; Fuchs et al., 2020; Geiger & Smidt, 2022) to balance symmetry-awareness with high-dimensional feature learning. While these multi-branch designs are promising, a systematic integration of parity-aware equivariant encoding and invariant graph-level classification for structured CAD data remains underexplored.

Summary and Positioning. In summary, while existing methods have made significant strides in 3D shape analysis and CAD processing (Guo et al., 2021; Jayaraman et al., 2021; Lambourne et al., 2021), they operate in distinct lanes: CAD-specific GNNs often overlook foundational 3D symmetries, while equivariant networks have not been adapted to the structured complexity of B-reps. EquiCAD provides a group-theoretic formulation for extracting $SO(3)/O(3)$ -equivariant B-rep features and uses invariant readout and graph reasoning for robust classification.

3. Methodology

3.1. Overall Framework

We present EquiCAD, a novel framework that systematically integrates $SO(3)/O(3)$ -equivariant learning with

graph-based reasoning (Gilmer et al., 2017; Wang et al., 2019) for robust 3D solid classification. Our key insight is that CAD models exhibit inherent geometric symmetries that should be preserved throughout the learning process, while simultaneously leveraging the structured topological information embedded in boundary representations (Jayaraman et al., 2021; Lambourne et al., 2021). As illustrated in Figure 1, EquiCAD comprises three core components: data representation, symmetry-aware feature fusion, and topology-aware graph aggregation.

Data Preprocessing and Representation. Each 3D solid is represented as a face-adjacency graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where nodes $\mathcal{V}$ correspond to parametric surfaces and edges $\mathcal{E}$ represent shared boundary curves. Each surface $s \in \mathcal{V}$ is uniformly sampled on a regular UV grid, producing a tensor of shape $(B_{\text{node}}, 7, H, W)$ containing 3D coordinates, normal vectors, and trimming masks. Similarly, curves $c \in \mathcal{E}$ are sampled to yield $(B_{\text{edge}}, 6, N_c)$ tensors encoding coordinates and tangents. Here B_{node} and B_{edge} denote the numbers of sampled surfaces and curves in a mini-batch, H and W are UV-grid dimensions, and N_c is the number of curve samples. All geometric data are normalized via bounding box normalization to ensure invariance to translation and scale.

Our graph is defined by B-rep topology rather than geo-

metric continuity. Each node corresponds to one topological face, and an edge is created only when two faces share an explicit boundary curve. Thus, G^1/C^1 continuity alone does not merge two faces: a smooth region is represented as two adjacent nodes if the B-rep stores it as two faces, and as one node if it is stored as a single face. Our benchmarks consist of valid manifold solid B-reps, where each boundary curve is incident to two faces. Handling open shells or non-manifold inputs with one or more than two incident faces would require modifying the incidence representation and is left for future work.

Symmetry-Aware Feature Encoding. The core innovation of EquiCAD lies in its dual-path encoding strategy. The *equivariant branch* processes geometric features using $SO(3)/O(3)$ -equivariant layers that explicitly preserve transformation properties under 3D rotations and reflections. The auxiliary *CNN branch* operates on invariantized equivariant features, preserving the final shape-level invariance while capturing local UV-space patterns.

Topology-aware Graph Aggregation. Encoded surface and curve features from both branches are fused via element-wise summation and then propagated through a message-passing graph neural network. This graph aggregation module models interactions between adjacent faces and curves, effectively capturing the complex topological relationships that characterize solid models. The final graph-level embedding is obtained through a multiscale readout mechanism that integrates information across all message-passing layers.

3.2. Symmetry-Aware Feature Encoding

The equivariant branch forms the mathematical foundation of our approach, ensuring that learned representations transform consistently with the input geometry under 3D rotations and reflections. We use the same notation for surface and curve encoders and take a surface $s \in \mathcal{V}$ with feature tensor $\mathbf{s}$ as an example.

Physical Parity and Irrep Typing. A fundamental contribution of EquiCAD is our principal group-theoretic formulation for the encoding of B-rep features. Unlike previous works that treat geometric attributes as generic feature vectors, we rigorously assign each attribute type to its corresponding irreducible representation based on its behavior under rotations and reflections.

The mathematical justification comes from the transformation behavior of differential geometric entities under orthogonal transformations. For a rotation or reflection $Q \in O(3)$, the transformation rules follow the physical parity of each geometric entity:

- **Curve Encoding** (“ $2 \times 1o$ ”): Both coordinates and

tangents transform as polar vectors ($l = 1, p = o$), since for any $Q \in O(3)$:

$$Q\mathbf{x} = \rho_1^o(Q)\mathbf{x}, \quad Q\mathbf{t} = \rho_1^o(Q)\mathbf{t} \quad (1)$$

where $\mathbf{x}$ denotes the position, $\mathbf{t}$ the tangent vector, and $\rho_l^p(Q)$ is the group representation acting on type (l, p) features.

- **Surface Encoding** (“ $1 \times 1o + 1 \times 1e + 1 \times 0e$ ”): We carefully distinguish between different transformation behaviors:

$$Q\mathbf{x} = \rho_1^o(Q)\mathbf{x} \quad \text{coordinates, polar} \quad (2)$$

$$\det(Q)Q\mathbf{n} = \rho_1^e(Q)\mathbf{n} \quad \text{normals, axial} \quad (3)$$

$$m = \rho_0^e(Q)m \quad \text{trimming mask, scalar} \quad (4)$$

where $\rho_1^e(Q)$ is the representation on axial vectors, $\rho_0^e(Q)$ is the trivial representation on scalars (identity action), and m is a binary trimming mask.

This careful typing is crucial to maintain physical consistency. The determinant factor $\det(Q)$ in the normal transformation explains why normals must be assigned even parity ($p = e$): they behave as pseudo-vectors under reflections. More detailed foundations are provided in Appendix A.

Mathematical Foundation. The mapping from Cartesian coordinates to irreducible representations follows the spherical harmonic expansion. For an input vector field $v(\mathbf{r})$, we compute its projection coefficient:

$$a_{li} = \int v(\mathbf{r}) Y_l^i(\theta, \phi) d\Omega, \quad (5)$$

where l is the harmonic degree, i is the order, Y_l^i is a real spherical harmonic, and $d\Omega$ is the unit-sphere surface element. Tensor products follow the Clebsch-Gordan decomposition:

$$(l_1, p_1) \otimes (l_2, p_2) = \bigoplus_{L=|l_1-l_2|}^{l_1+l_2} (L, p_1 \cdot p_2). \quad (6)$$

where $p_1 \cdot p_2$ denotes parity multiplication. For curve encoding, two polar vectors yield:

$$(1, o) \otimes (1, o) = (0, e) \oplus (1, e) \oplus (2, e). \quad (7)$$

The above equation describes how vector inputs naturally produce scalar and higher-order tensor components. This mathematical foundation ensures that all feature transformations respect the $O(3)$ symmetry group.

Equivariant Network Architecture. Our architecture preserves $SO(3)/O(3)$ -equivariance through tensor product

operations. The equivariant linear layer between input type τ_{in} and output type τ_{out} is:

$$\mathbf{z} = \text{Linear}_{\tau_{\text{in}} \rightarrow \tau_{\text{out}}}(\mathbf{s}) = \bigoplus_j (W_j \otimes I_{2l_j+1}) \mathbf{s}^{(j)}. \quad (8)$$

Here $\mathbf{s}^{(j)} \in \mathbb{R}^{m_{\text{in}}^{(j)} \times (2l_j+1)}$ is the j -th irrep block, $W_j \in \mathbb{R}^{m_{\text{out}}^{(j)} \times m_{\text{in}}^{(j)}}$ mixes only multiplicity channels, and I_{2l_j+1} is the identity over irrep components. Tensor product layers further enrich feature interactions:

$$\rho_{(L,p)}(g)(\mathbf{x} \otimes \mathbf{y}) = (\rho_{(l_1,p_1)}(g)\mathbf{x}) \otimes (\rho_{(l_2,p_2)}(g)\mathbf{y}). \quad (9)$$

Here $g \in O(3)$, $\mathbf{x}$ and $\mathbf{y}$ have types (l_1, p_1) and (l_2, p_2) , and (L, p) is one output irrep in their tensor product.

A critical design choice is that the final equivariant layer produces only scalar ($0e$) output, enabling spatial pooling to obtain $SO(3)/O(3)$ -invariant embeddings:

$$h_{s,\text{enn}} = \frac{1}{N} \sum_{i=1}^N y_i, \quad (10)$$

where y_i is the scalar output at sampling point i , and N is the number of points. We obtain the curve embedding $h_{c,\text{enn}}$ analogously with separate weights.

3.3. Multi-Branch Feature Fusion

While the equivariant branch enforces symmetry-preserving transformations, we complement it with conventional CNNs to capture high-capacity local patterns.

Invariant Auxiliary CNN Branch. The CNN branch does not operate on pose-dependent raw geometric tensors. Instead, it takes invariantized features derived from the equivariant representation: each non-scalar irrep copy is converted to its Euclidean norm, while scalar $0e$ channels are kept unchanged. This produces invariant CNN embeddings $h_{s,\text{cnn}}$ and $h_{c,\text{cnn}}$ for each surface $s \in \mathcal{V}$ and curve $c \in \mathcal{E}$.

Feature Fusion Strategy. Both branches produce invariant embeddings before fusion:

$$h_s = h_{s,\text{cnn}} + h_{s,\text{enn}}, \quad s \in \mathcal{V}, \quad (11)$$

$$h_c = h_{c,\text{cnn}} + h_{c,\text{enn}}, \quad c \in \mathcal{E}, \quad (12)$$

$$H_s = \{h_s \mid s \in \mathcal{V}\}, \quad H_c = \{h_c \mid c \in \mathcal{E}\}. \quad (13)$$

Thus, the element-wise summation preserves $SO(3)/O(3)$ -invariance up to floating-point error.

The fusion occurs before graph aggregation, ensuring that both symmetry-aware and data-driven features participate in the topological reasoning process.

3.4. Topology-Aware Graph Aggregation

The graph aggregation module models the complex relationships between geometric entities in B-rep models while maintaining the learned symmetry properties.

Message Passing Formulation. We initialize node features with surface embeddings H_s and edge features with curve embeddings H_c . The message passing follows a multi-layer architecture with residual connections. The edge update is:

$$e_{uv}^{(l+1)} = M_E \left((1 + \epsilon_E) e_{uv}^{(l)} + M_P(h_u^{(l)}) + M_P(h_v^{(l)}) \right), \quad (14)$$

where $e_{uv}^{(l)}$ is the edge feature at layer l , M_E and M_P are MLPs, ϵ_E is learnable, and $h_u^{(l)}, h_v^{(l)}$ are node features at layer l .

The node update employs an edge-conditioned convolution:

$$\begin{aligned} \tilde{h}_u^{(l+1)} &= \sum_{v \in \mathcal{N}(u)} W_{uv} h_v^{(l)}, \\ h_u^{(l+1)} &= M_N \left((1 + \epsilon_N) h_u^{(l)} + \tilde{h}_u^{(l+1)} \right), \end{aligned} \quad (15)$$

where $\mathcal{N}(u)$ is the neighbor set, $W_{uv} \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ is generated from edge features, M_N is an MLP, ϵ_N is learnable, and $d_{\text{in}}, d_{\text{out}}$ are feature dimensions.

Multi-Scale Graph Readout. After T message-passing layers, we compute the graph-level embedding using jumping-knowledge aggregation:

$$H_G = \sum_{k=0}^T \text{Dropout} \left(\text{Linear}_k \left(\text{POOL}(h^{(k)}) \right) \right), \quad (16)$$

where $h^{(k)}$ is the node feature at layer k , Linear_k is a layer-specific projection, and POOL is permutation-invariant pooling. The resulting graph embedding is invariant to both $O(3)$ transformations and arbitrary face ordering. We optimize the network with cross-entropy loss.

4. Experiments

4.1. Datasets

We evaluate EquiCAD on four benchmarks: SolidLetters, Parts, Machining Feature, and Features. **SolidLetters** contains 96,000 3D letter shapes with rotational symmetries, serving as a controlled testbed for symmetry-aware methods. **Parts** comprises 15,000 industrial mechanical components with complex topological structures, representing real-world CAD challenges. **Machining Feature** is a multi-class CAD benchmark with 24,000 models, where each solid is assigned a machining-feature category (Zhang



Figure 2. Representative examples from the Features dataset, showcasing diverse geometric features including through slots, blind steps, pockets, and chamfers.

et al., 2018). In contrast, the **Features** dataset is our multi-task binary benchmark for fine-grained feature recognition constructed in this work.

Features Dataset Construction (Multi-task Binary): We build this benchmark by repurposing a segmentation-style B-rep corpus with machining-feature annotations (BrepMFR) (Zhang et al., 2024) into 24 binary classification tasks. Each sub-dataset targets a specific geometric feature type (e.g., slots, pockets, chamfers). For each feature type c , we extract positive samples from shapes containing annotated regions of type c and select negative samples from shapes completely lacking that feature. We then rank all shapes by geometric-topological complexity using multiple metrics and select the 2,000 most complex positives and 2,000 most complex negatives for the final dataset. The main table reports the 16 challenging and stable tasks; the remaining 8 tasks saturate near AUROC 1.0 for most baselines. Figure 2 shows representative examples.

Data Preprocessing: All models undergo consistent preprocessing: we parse B-rep structures, reconstruct face-edge graphs, and apply bounding-box normalization. Faces are sampled on 16×16 UV grids (7 channels: coordinates, normals, trimming mask), while edges are uniformly sampled into 64-point curves (6 channels: coordinates, tangents).

4.2. Baseline Methods

We benchmark against two families of methods to cover both unstructured and CAD-structured settings. Point-based baselines include PointNet (Qi et al., 2017a), PointNet++ (Qi et al., 2017b), DGCNN (Wang et al., 2019),

and Vector Neurons (VN) (Deng et al., 2021): PointNet provides permutation-invariant set processing; PointNet++ adds hierarchical set abstraction; DGCNN forms dynamic k NN graphs; and VN enables $SO(3)$ -equivariant learning on point clouds. CAD-oriented baselines operate directly on B-reps: UV-Net (Jayaraman et al., 2021) combines UV-domain CNN encoders with a face-edge GNN; AAGNet (Wu et al., 2024) propagates geometric/topological cues on B-rep graphs; DFGAT (Park et al., 2025) uses graph attention over B-rep-derived descriptors; and DTGBrepGen (Li et al., 2025) incorporates topology-aware constraints and geometric signals beyond UV-only designs.

For protocol and fairness, we use the train/validation/test splits defined in Section 4.3 and select checkpoints by validation loss. CAD methods receive B-rep UV tensors and face-edge graphs produced by the same preprocessing pipeline; point-based methods are fed uniformly sampled points from the same shapes with identical normalization. Results are reported as the mean over three runs unless stated otherwise.

4.3. Implementation Details

Experimental Setup: We employ fixed dataset splits of 16:4:5 for SolidLetters and 8:1:1 for Parts, Machining Feature, and Features. For Features, splits are applied independently per sub-dataset. EquiCAD is trained with AdamW, learning rate 3×10^{-4} , weight decay 10^{-4} , batch size 64, and cosine annealing on a single NVIDIA RTX 3090 GPU. We train for up to 100 epochs with validation-loss-based early stopping and select the checkpoint with the lowest validation loss. We use no additional data augmentation beyond centering and bounding-box normalization. Results are averaged over three runs with random seeds de-

Method	SolidLetters (%)	Parts (%)	Machining Feature (%)	Params (M)
UV-Net (Jayaraman et al., 2021)	96.96 $\pm$ 0.15	86.80 $\pm$ 0.10	99.74 $\pm$ 0.10	1.34
PointNet (Qi et al., 2017a)	96.92 $\pm$ 0.09	81.55 $\pm$ 0.23	88.56 $\pm$ 0.12	0.81
PointNet++ (Qi et al., 2017b)	97.23 $\pm$ 0.13	88.62 $\pm$ 0.11	91.38 $\pm$ 0.14	1.42
DGCNN (Wang et al., 2019)	94.93 $\pm$ 0.17	74.44 $\pm$ 0.33	93.42 $\pm$ 0.33	1.81
Vector Neurons (Deng et al., 2021)	95.43 $\pm$ 0.24	82.16 $\pm$ 0.15	92.96 $\pm$ 0.17	2.10
AAGNet (Wu et al., 2024)	95.19 $\pm$ 0.06	90.53 $\pm$ 0.09	98.44 $\pm$ 0.12	0.38
DFGAT (Park et al., 2025)	95.93 $\pm$ 0.12	90.30 $\pm$ 0.07	98.54 $\pm$ 0.09	0.61
DTGBrepGen (Li et al., 2025)	95.54 $\pm$ 0.20	90.59 $\pm$ 0.13	98.73 $\pm$ 0.37	34.26
EquiCAD (ours)	97.64$\pm$0.05	91.66$\pm$0.14	99.87$\pm$0.01	1.36

Table 1. Performance (%), mean $\pm$ std over 3 runs) on SolidLetters, Parts, and Machining Feature (multi-class) benchmarks, along with the number of parameters.

terminated by the training start time. Approximate training time per seed is 9 h on SolidLetters, 5 h on Parts, 2 h on Machining Feature, and 0.75 h on each Features task.

EquiCAD Architecture: Our framework combines $SO(3)/O(3)$ -equivariant and invariant auxiliary CNN branches with the following settings. **Equivariant Branch:** Curve encoder: “ $2 \times 1o \rightarrow 8 \times 0e + 8 \times 2e \rightarrow 64 \times 0e$ ”; surface encoder: “ $1 \times 1o + 1 \times 1e + 1 \times 0e \rightarrow 8 \times 0e + 8 \times 1o + 8 \times 1e + 4 \times 2e \rightarrow 64 \times 0e$ ”. **CNN Branch:** 1D and 2D CNNs with hidden channels (64, 128, 256) are applied to invariantized curve and surface features as described in Section 3.3. **Graph Aggregation:** a 3-layer GNN with 64-dimensional hidden states. For SolidLetters, Parts, and Machining Feature, we report mean $\pm$ std classification accuracy over three runs. For Features, we report mean AUROC over three runs.

4.4. Main Results

SolidLetters, Parts, and Machining Feature Benchmarks: Table 1 summarizes classification accuracy (mean $\pm$ std over 3 runs) and model size. On SolidLetters, EquiCAD achieves 97.64%, outperforming PointNet++ (97.23%) and DFGAT (95.93%). On Parts, EquiCAD reaches 91.66%, improving over DFGAT (90.30%), DTGBrepGen (90.59%), and AAGNet (90.53%). On Machining Feature, EquiCAD obtains 99.87%, surpassing UV-Net (99.74%) with a comparable number of parameters.

Features Dataset Analysis (Multi-task Binary): Table 2 presents AUROC scores for 16 challenging feature recognition tasks. EquiCAD achieves the highest performance in the majority of tasks, with strong gains on triangular passages (0.998 vs 0.859) and rectangular pockets (1.000 vs 0.839). Averaged over all 24 Features tasks, EquiCAD remains best with 0.807 AUROC, compared with 0.742 for the strongest baseline; the full 24-task average is reported in Appendix Table 5.

4.5. Ablation Study

We systematically evaluate the contribution of each component via the ablation study in Table 3, consisting of seven variants. **CNN-only:** Remove equivariant encoders from both curve and surface branches, leaving only the CNN encoders. **EquiNN-only:** Remove the CNN auxiliary encoders from both branches, leaving only the $SO(3)/O(3)$ -equivariant encoders. **Ours-Curve_CNN:** Remove the equivariant encoder from the curve branch (CNN-only curves), while keeping the surface branch as the full configuration (equivariant encoder + CNN auxiliary). **Ours-Surface_CNN:** Remove the equivariant encoder from the surface branch (CNN-only surfaces), while keeping the curve branch as the full configuration. **Ours-Curve_EquiNN:** Remove the CNN auxiliary from the curve branch (equivariant-only curves), while keeping the surface branch as the full configuration. **Ours-Surface_EquiNN:** Remove the CNN auxiliary from the surface branch (equivariant-only surfaces), while keeping the curve branch as the full configuration. **EquiCAD (full):** Both branches use the equivariant encoder together with the CNN auxiliary (our complete model). As shown in Table 3, the full model achieves the best performance on both datasets. Importantly, the two new variants (Ours-Curve_EquiNN and Ours-Surface_EquiNN) show that removing the CNN auxiliary from either branch still degrades performance, indicating that the CNN auxiliary provides complementary high-capacity features beyond the symmetry-aware equivariant representations.

4.6. Robustness Analysis

Table 4 directly evaluates transformation robustness on Parts. Rotations use $60^\circ, 120^\circ, 180^\circ, 240^\circ, 300^\circ$ around axis $(1, 1, 1)/\sqrt{3}$, and reflection uses the plane normal $(1, 1, 1)/\sqrt{3}$. Consistency measures agreement with predictions on the original input. EquiCAD remains stable un-

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Feature Type	PointNet	PointNet++	DGCNN	UV-Net	AAGNet	DTGBrepGen	EquiCAD (ours)
Rectangular through slot	0.538	0.570	0.565	0.615	0.513	0.510	0.706
Triangular through slot	0.542	0.567	0.524	0.858	0.537	0.794	0.982
Circular through slot	0.569	0.580	0.503	0.510	0.522	0.470	0.594
Triangular passage	0.537	0.535	0.483	0.859	0.591	0.515	0.998
Rectangular through step	0.589	0.571	0.524	0.536	0.556	0.506	0.583
2-Sided through step	0.541	0.572	0.548	0.512	0.546	0.513	0.594
Slanted through step	0.553	0.570	0.588	0.518	0.546	0.479	0.579
Rectangular blind step	0.544	0.583	0.599	0.570	0.494	0.512	0.843
Triangular blind step	0.561	0.570	0.547	0.823	0.521	0.537	0.979
Circular blind step	0.543	0.593	0.525	0.557	0.501	0.510	0.604
Rectangular blind slot	0.541	0.514	0.525	0.495	0.520	0.527	0.572
Horizontal circular end blind slot	0.517	0.570	0.554	0.509	0.531	0.509	0.570
Vertical circular end blind slot	0.561	0.581	0.529	0.549	0.503	0.517	0.590
Rectangular pocket	0.562	0.573	0.544	0.839	0.525	0.510	1.000
Chamfer	0.518	0.519	0.559	0.575	0.518	0.511	0.615
Fillet	0.568	0.598	0.582	0.476	0.551	0.511	0.550
Average	0.549	0.567	0.544	0.613	0.530	0.527	0.710

Table 2. AUROC performance on 16 challenging binary feature recognition tasks (mean over 3 runs). Best results in bold.

Configuration	SolidLetters	Parts
CNN-only	96.96%	86.70%
EquiNN-only	95.27%	88.03%
Ours-Curve_CNN	96.12%	91.32%
Ours-Surface_CNN	96.33%	91.05%
Ours-Curve_EquiNN	95.57%	90.52%
Ours-Surface_EquiNN	95.40%	90.26%
EquiCAD (full)	97.64%	91.66%

Table 3. Ablation study.

der both rotations and reflections, while DFGAT degrades sharply.

We further evaluate sensitivity to irrep choices in Appendix A.4. Mistyping surface normals as polar vectors causes substantial degradation, validating the physical parity assignment.

5. Limitations

EquiCAD introduces additional computational cost compared with lightweight B-rep GNNs because equivariant tensor operations process typed geometric features before graph aggregation, although the full model remains compact with 1.36M parameters. Like other B-rep-based methods, it assumes clean CAD inputs with valid topology, reliable UV parameterization, and well-defined normals; corrupted topology, missing or trimmed faces, noisy nor-

mals, or non-manifold/open-shell inputs may degrade performance. Finally, the Features benchmark is designed as a challenging balanced stress test for fine-grained feature recognition, so datasets with different feature taxonomies or CAD modeling conventions may require label mapping or re-annotation.

6. Conclusion

We presented EquiCAD, a framework that integrates $SO(3)/O(3)$ -equivariant learning with graph neural networks for 3D solid classification. Its key contribution is a symmetry-aware encoding scheme that decomposes B-rep entities into irreducible representations based on physical parity and converts them to invariant embeddings before classification. Experiments across SolidLetters, Parts, Machining Feature, and Features demonstrate state-of-the-art performance, strong transformation robustness, and improved fine-grained feature recognition. Future work will explore applications to other CAD tasks and broader out-of-distribution evaluation.

Software and Data

We will release the full codebase, preprocessing scripts, processed data resources for non-commercial research use, and train/validation/test splits after publication.

Model	Orig. Acc.	Avg. Rot. Acc.	Refl. Acc.	Avg. Rot. Cons.	Refl. Cons.
EquiCAD	0.9159	0.9159	0.9159	1.0000	1.0000
DFGAT	0.9030	0.3615	0.1934	0.3785	0.2109

Table 4. Transformation robustness on Parts.

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Impact Statement

This paper presents work whose goal is to advance machine learning for robust CAD understanding. We anticipate primarily positive impacts through improved reliability in downstream design and manufacturing workflows. We do not foresee specific negative societal consequences that require separate discussion here.

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A. Appendix

A.1. Preliminaries: $SO(3)/O(3)$ -Equivariant Deep Learning

To ensure our framework is accessible to readers unfamiliar with geometric deep learning, we briefly review the essential concepts of $SO(3)/O(3)$ -equivariant neural networks. Our focus is on the orthogonal group $O(3)$ —the group of 3D rotations and reflections—which provides the mathematical foundation for handling the symmetries inherent in CAD models.

$SO(3)/O(3)$ -Equivariance Principle. A function $f : X \rightarrow Y$ is *equivariant* to a group G if transforming the input by any group element $g \in G$ results in a predictable transformation of the output:

$$f(\rho_X(g) \cdot x) = \rho_Y(g) \cdot f(x), \quad \forall g \in G, \quad (17)$$

where ρ_X and ρ_Y are representations of G on the input and output spaces, respectively. In our context, $G = O(3)$, and equivariance ensures that rotating or reflecting a B-rep model consistently transforms its learned features—a crucial property for robust CAD analysis.

Irreducible Representations of $SO(3)/O(3)$. To build $SO(3)/O(3)$ -equivariant networks, we decompose features into *irreducible representations* (irreps)—the fundamental building blocks that transform in a simple, predictable way under group actions. Each irrep is characterized by an integer *degree* $l \geq 0$ and a *parity* $p \in \{e, o\}$:

- Degree l determines the dimensionality ($2l + 1$) and angular frequency of the transformation behavior.
- Parity p specifies behavior under reflection: *even* ($p = e$) features remain unchanged, while *odd* ($p = o$) features change sign.

For example, scalar values ($l = 0, p = e$) are invariant to rotations, while polar vectors (e.g., coordinates) transform as $l = 1, p = o$ irreps.

Feature Encoding with Multiplicities. Practical $SO(3)/O(3)$ -equivariant networks compose features as direct sums of irreps. For each irrep type (l, p) , we can include multiple independent copies (multiplicity m). Thus, a feature vector containing K distinct irrep types has total dimension:

$$D = \sum_{i=1}^K m_i \cdot (2l_i + 1). \quad (18)$$

Following conventions in libraries like e3nn (Geiger & Smidt, 2022), we notationally express feature types as concatenations like $"2 \times 1o + 3 \times 0e"$, denoting two vector-valued ($l = 1$, odd) and three scalar ($l = 0$, even) channels.

$SO(3)/O(3)$ -Equivariant Linear Maps and Nonlinearities. $SO(3)/O(3)$ -Equivariant linear layers between input and output features of types ρ_{in} and ρ_{out} must satisfy the intertwining condition:

$$W \cdot \rho_{\text{in}}(g) = \rho_{\text{out}}(g) \cdot W, \quad \forall g \in O(3). \quad (19)$$

Such constraints drastically reduce the number of free parameters in W , improving generalization. For nonlinearities, standard activation functions (e.g., ReLU) generally break equivariance. Instead, we use *gate nonlinearities* (Weiler et al., 2018) that combine tensor products of irreps with learned gating scalars, preserving equivariance while enabling complex feature learning.

These foundations enable our key contribution: adapting $SO(3)/O(3)$ -equivariance to the structured graph domain of CAD B-reps, building upon established graph neural network principles (Gilmer et al., 2017; Kipf & Welling, 2017) while incorporating geometric symmetry constraints.

A.2. Features Benchmark: Sub-task Construction and Selection

The Features dataset is built from a segmentation-style B-rep corpus with machining-feature annotations (BrepMFR) (Zhang et al., 2024), which we repurpose into 24 binary feature-recognition tasks. Each sub-dataset targets one feature type (e.g., slots, pockets, chamfers). For a given feature type c , we construct a binary classification task by treating shapes that contain an annotated region of type c as positive samples and shapes that entirely lack feature type c as negative samples. To emphasize challenging cases, we additionally rank shapes by geometric-topological complexity using multiple metrics and keep the most complex positives and negatives per task.

Although we initially defined 24 candidate tasks, we report results on 16 sub-tasks in the main benchmark. The remaining 8 tasks were excluded because they are too easy, with AUROC saturating close to 1.0 across diverse base-lines, thus offering limited discriminatory value. Focusing on the 16 challenging and stable tasks yields a more meaningful evaluation of fine-grained feature recognition.

More broadly, this benchmark is designed to fill a gap in existing CAD learning datasets: many prior benchmarks emphasize global shape classification or retrieval, whereas the Features dataset stresses semantically grounded, feature-level recognition from B-rep representations. By providing a suite of controlled binary tasks derived from a shared B-rep corpus, it enables systematic evaluation of fine-grained geometric-topological understanding.

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Model	PointNet	PointNet++	DGCNN	UV-Net	AAGNet	DTGBrepGen	EquiCAD
Average AUROC	0.699	0.711	0.696	0.742	0.687	0.685	0.807

Table 5. Average AUROC over all 24 Features tasks.

Model	Setting	Accuracy	Macro-F1
EquiCAD	Original input	0.9159	0.8192
EquiCAD	Rotation TTA	0.9159	0.8192
EquiCAD	All-transform TTA	0.9159	0.8192
DFGAT	Original input	0.9030	0.8179
DFGAT	Rotation TTA	0.5218	0.3225
DFGAT	All-transform TTA	0.4739	0.2794

Table 6. Test-time augmentation results on Parts. Average rotation/reflection JSD values are 0.000000/0.000000 for EquiCAD and 0.365398/0.476381 for DFGAT.

Normal noise std.	Feature 1 AUROC	Feature 2 AUROC
0.00	0.7267	0.9871
0.05	0.7269	0.9874
0.10	0.7267	0.9877
0.15	0.7259	0.9875
0.20	0.7254	0.9876

Table 7. Corrupted-input robustness under Gaussian noise injected into surface-normal channels on the Features benchmark.

Curve Irreps	Surface Irreps	SolidLetters	Parts
$2 \times 1o$	$1 \times 1o + 1 \times 1e + 1 \times 0e$	97.64%	91.66%
$2 \times 1o$	$2 \times 1o + 1 \times 0e$	42.49%	59.37%
$2 \times 1o$	$1 \times 1o + 4 \times 0e$	96.16%	91.22%
$2 \times 1o$	$7 \times 0e$	94.96%	91.19%
$1 \times 1o + 3 \times 0e$	$2 \times 1o + 1 \times 0e$	95.53%	91.09%
$1 \times 1o + 3 \times 0e$	$1 \times 1o + 4 \times 0e$	95.67%	88.46%
$1 \times 1o + 3 \times 0e$	$7 \times 0e$	94.84%	91.19%
$6 \times 0e$	$2 \times 1o + 1 \times 0e$	96.89%	88.70%
$6 \times 0e$	$1 \times 1o + 4 \times 0e$	96.87%	90.48%
$6 \times 0e$	$7 \times 0e$	96.89%	90.98%
$1 \times 1o + 3 \times 0e$	$1 \times 1o + 1 \times 1e + 1 \times 0e$	95.30%	90.63%
$6 \times 0e$	$1 \times 1o + 1 \times 1e + 1 \times 0e$	95.37%	90.89%
$2 \times 1o$	$2 \times 1e + 1 \times 0e$	95.43%	91.05%
$1 \times 1o + 3 \times 0e$	$2 \times 1e + 1 \times 0e$	95.48%	90.46%
$6 \times 0e$	$2 \times 1e + 1 \times 0e$	95.16%	90.59%

Table 8. Comprehensive robustness analysis: combining main paper results with additional configurations.

A.3. Additional Transformation and Corruption Robustness

Table 6 reports the Jensen-Shannon divergence (JSD) and test-time augmentation (TTA) results for the Parts transformation stress test. For EquiCAD, TTA yields the same result as the original input, while DFGAT remains substantially below its clean-input accuracy.

A.4. Complete Robustness Experiments

In this subsection, we present a comprehensive robustness analysis table that combines results shown in the main paper with additional configurations not included due to space constraints. This extended table provides a full view of the performance under various irreducible representation choices and physical parity assignments.